

# SELMER ELLIPTIC CURVES

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This worksheet studies a single family of plane cubics and uses it to reconstruct, by hand, some of the main results of [Sil09, §§III.1–III.4]: Weierstrass equations, the group law, the Riemann–Roch construction of a Weierstrass model, and isogenies. The same family is worked out in [Section 4.8](#) of the notes [AP26], in the same notation; the problems here are the computations that section summarizes, so do them before reading it. Sections 1–2 are meant for the problem session; Section 3 may be started there and finished at home; Section 4 is a set of worked Magma examples to read and re-run afterwards, with three short variations to try at the end.

As a rough guide, allow about 12 minutes for §1 and 33 for §2. Within §2, Problem 2.2 is the core and Problem 2.5(e)–(g) the payoff; if time is short, leave Problem 2.5(a)–(d) and Problem 2.6 for later. Section 4 needs no time in the session; it is worked examples, not exercises.

**Standing notation.** Throughout,  $k$  is a field with  $\text{char } k \neq 3$ ,  $\bar{k}$  is an algebraic closure of  $k$ , and  $d \in k^\times$ . We write

$$F := X^3 + Y^3 - dZ^3, \quad C_d := \{F = 0\} \subset \mathbb{P}_k^2, \quad O := (1 : -1 : 0),$$

and  $\zeta \in \bar{k}$  for a primitive cube root of unity. A pair  $(C_d, O)$  of this shape is called a [Selmer elliptic curve](#), after [Sel51]; Section 5 explains the name. As usual,  $X, Y, Z$  are projective coordinates and lowercase letters are affine coordinates. We write  $\overline{C}_d := C_d \times_k \bar{k}$ .

## 1. WARM-UP: A PLANE CUBIC AND A RATIONAL POINT ON IT

**Problem 1.1** (It is an elliptic curve).

Recall from [Sil09, §III.3], or [Definition 4.1.1](#) of the notes, that an elliptic curve over  $k$  is a pair  $(E, O)$  with  $E$  a smooth, projective, geometrically irreducible curve of genus one over  $k$  and  $O \in E(k)$  a fixed rational point.

- Check that  $O \in C_d(k)$ .
- Compute the three partial derivatives of  $F$  and deduce that  $C_d$  is smooth, using  $\text{char } k \neq 3$  and  $d \neq 0$ . Where is each hypothesis used?
- Suppose instead that  $\text{char } k = 3$  (only for this problem). Show that over  $\bar{k}$  the curve  $C_d$  is a *triple line*.<sup>a</sup>
- Show that a smooth plane curve is geometrically irreducible.<sup>b</sup>
- Conclude from the degree–genus formula that  $(C_d, O)$  is an elliptic curve over  $k$ .

<sup>a</sup>Frobenius.

<sup>b</sup>Bézout.

**Problem 1.2** (The base point is an inflection point).

- Show that the line  $Z = 0$  meets  $C_d$  in three *distinct* points of  $C_d(\bar{k})$ , and write them down. For which  $k$  are all three in  $C_d(k)$ ?

- (b) Compute the tangent line  $L$  to  $C_d$  at  $O$ , and show that  $L \cap C_d = 3O$ . (So  $O$  is an **inflection point**, or **flex**.)
- (c) In a Weierstrass model, which line plays the role of  $L$ ? What is different here?
- (d) Let  $\oplus$  denote the group law on  $C_d$  with identity  $O$ , defined by the chord–tangent construction of [Sil09, §III.2].<sup>a</sup> Using (b), show that for  $P, Q, R \in C_d(\bar{k})$ ,

$$P \oplus Q \oplus R = O \iff P + Q + R \text{ is cut out on } C_d \text{ by a line.}$$

<sup>a</sup>[Sil09, Prop. III.3.4] identifies  $\oplus$  with addition in  $\text{Pic}^0 \bar{C}_d$  via  $P \mapsto [P - O]$ .

## 2. THE GROUP LAW

**Problem 2.1** (Negation, and 2-torsion).

- (a) Show that  $\ominus(X_0 : Y_0 : Z_0) = (Y_0 : X_0 : Z_0)$  for every point  $(X_0 : Y_0 : Z_0) \in C_d(\bar{k})$ . There are two things to check: that the right-hand side lies on  $C_d$ , and that it is the third intersection point of  $C_d$  with the line through  $O$  and  $(X_0 : Y_0 : Z_0)$ .
- (b) Deduce that the nontrivial 2-torsion of  $C_d$  is cut out by the line  $X = Y$ , and that  $\#C_d[2](\bar{k}) = 4$ .
- (c) Take  $k = \mathbb{Q}$ . For which  $d$  does  $C_d$  have a nontrivial rational 2-torsion point? Write it down, and give the smallest positive integer  $d$  for which this happens.

Part (b) of Problem 2.2 is its heart; parts (d) and (e) are then mechanical. If time is short, take (b) on faith and come back to its proof, but do not skip (c): it is what makes (e) painless. The expansion identity is the polar expansion of Lemma 4.2.5 in the notes, in the one case where its coefficients come from a trilinear form; the formulas of (d) and (e) are Proposition 4.8.8 there.

**Problem 2.2** (The trilinear form, and the addition formulas).

For vectors  $P = (X_P, Y_P, Z_P)$ ,  $Q$  and  $R$  in  $\bar{k}^3$ , define

$$(2.1) \quad \langle P, Q, R \rangle := X_P X_Q X_R + Y_P Y_Q Y_R - d Z_P Z_Q Z_R.$$

- (a) Check that (2.1) is symmetric in  $P, Q, R$ , linear in each argument separately, and satisfies  $\langle P, P, P \rangle = F(P)$ .
- (b) Prove the *expansion identity*: for scalars  $s, t$ ,

$$(2.2) \quad F(sP + tQ) = s^3 \langle P, P, P \rangle + 3s^2 t \langle P, P, Q \rangle + 3st^2 \langle P, Q, Q \rangle + t^3 \langle Q, Q, Q \rangle.$$

Where do the 3's come from?<sup>a</sup>

- (c) Prove that  $3\langle P, P, Q \rangle = \nabla F(P) \cdot Q$ , and deduce that for  $P \in C_d(\bar{k})$ ,

$$\langle P, P, Q \rangle = 0 \iff Q \text{ lies on the tangent line to } C_d \text{ at } P.$$

- (d) Let  $P_1 = (X_1 : Y_1 : Z_1)$  and  $P_2 = (X_2 : Y_2 : Z_2)$  be distinct points of  $C_d(\bar{k})$ , and parametrise the line through them as  $sP_1 + tP_2$ . Using (b), show that the third intersection point is at  $(s : t) = (B : -A)$  where  $A = \langle P_1, P_1, P_2 \rangle$  and  $B = \langle P_1, P_2, P_2 \rangle$ , and hence that

$$(2.3) \quad P_1 \oplus P_2 = (BY_1 - AY_2 : BX_1 - AX_2 : BZ_1 - AZ_2).$$

Why can  $A$  and  $B$  not both vanish?

- (e) Let  $P = (X_0 : Y_0 : Z_0) \in C_d(\bar{k})$  and put  $Q := (Y_0^2 : -X_0^2 : 0)$ . Verify that  $Q$  lies on the tangent to  $C_d$  at  $P$ , and use (b) and (c) to derive

$$(2.4) \quad [2]P = (-Y_0(2X_0^3 + Y_0^3) : X_0(X_0^3 + 2Y_0^3) : Z_0(X_0^3 - Y_0^3)).$$

- (f) Check (2.4) against Problem 2.1: what does it give on the line  $X = Y$ ? (The line  $X = 0$  is taken up in Problem 2.5.)
- (g) Where does the whole method break down if  $\text{char } k = 3$ ? Compare your answer with Problem 1.1(c).

<sup>a</sup>Expand  $\langle sP + tQ, sP + tQ, sP + tQ \rangle$  by trilinearity into eight terms and collect using symmetry.

**Problem 2.3** (Arithmetic on  $C_7$ ).

Let  $d = 7$ , so that  $P := (2 : -1 : 1) \in C_7(\mathbb{Q})$  because  $2^3 + (-1)^3 = 7$ .

- (a) Compute  $[2]P$  using (2.4), and simplify. Verify your answer by exhibiting an identity between cubes of integers.
- (b) Compute  $[3]P$  using (2.3) applied to  $P$  and  $[2]P$ , and verify it the same way.
- (c) Compute  $[4]P$ .
- (d) The three rational points you have found have numerators and denominators of rapidly increasing size. What does this suggest about the order of  $P$  in  $C_7(\mathbb{Q})$ ? (You are not asked to prove it; see [Sil09, §VIII.9] or [HS00, Part B].)

**Fact 2.4** (The Hessian criterion). Let  $C = V(F) \subset \mathbb{P}_k^2$  be a smooth plane curve of degree  $m$  over a field of characteristic 0 or  $> 3$ , and let

$$H(F) = \det \left( \frac{\partial^2 F}{\partial X_i \partial X_j} \right)_{0 \leq i, j \leq 2}.$$

Then a point  $P \in C(\bar{k})$  is an inflection point if and only if  $H(F)(P) = 0$ . This is classical, and it genuinely fails in characteristic three; see Problem 2.5(h) below.

In Problem 2.5, parts (a)–(c) are the general theory and (e)–(g) the payoff for our family; if time is short, take (c) on faith and go to (e)–(g). Part (d) is the most satisfying: two completely different counts turn out to agree. The nine flexes are listed in Proposition 4.8.4 of the notes.

**Problem 2.5** (Inflection points and 3-torsion).

- (a) Show that  $H(F)$  is a form of degree  $3(m - 2)$ , so of degree 3 when  $m = 3$ . Assuming  $C$  is a smooth plane cubic, explain why  $C \not\subseteq V(H(F))$ , and conclude from Bézout [Har77, §I.7] that  $C$  has exactly 9 inflection points *counted with multiplicity*.
- (b) Now forget the Hessian. Let  $C$  be a smooth plane cubic over  $\bar{k}$  and suppose  $O \in C(\bar{k})$  is a flex, so that  $(C, O)$  is an elliptic curve with the group law of Problem 1.2(d). Show that

$$P \text{ is a flex} \iff [3]P = O,$$

so that the set of flexes is exactly  $C[3](\bar{k})$ .

- (c) Show that a smooth plane cubic over  $\bar{k}$  always *has* a flex, without using the Hessian.<sup>a</sup> Deduce that, with no flex singled out, the set of flexes is a torsor under  $C[3]$ ; in particular the number of flexes is  $\#C[3](\bar{k})$ .
- (d) Combine (b) and (c) with the structure of  $C[3]$  in characteristic  $\neq 3$  to conclude that a smooth plane cubic has exactly 9 flexes. What does comparing this with (a) tell you about the intersection  $C \cap V(H(F))$ ?
- (e) Specialise to  $C_d$ . Compute  $H(F)$  for  $F = X^3 + Y^3 - dZ^3$  and deduce that, as subsets of  $C_d(\bar{k})$ ,

$$C_d[3](\bar{k}) = (C_d \cap \{XYZ = 0\})(\bar{k}).$$

Write down all nine flexes.

- (f) Check (e) against the group law directly: use (2.4) to show that every point  $P = (0 : Y_0 : Z_0) \in C_d(\bar{k})$  satisfies  $[2]P = \ominus P$ .
- (g) Which of the nine flexes lie in  $C_d(k)$ ? Treat the three lines separately, and say what happens for  $k = \mathbb{Q}$ .
- (h) (★) In characteristic three,  $\#C[3](\bar{k})$  is 3 or 1 according as  $C$  is ordinary or supersingular, so a smooth plane cubic has 3 or 1 flexes. How is that compatible with (a)?

<sup>a</sup>For any  $P_0 \in C(\bar{k})$  the class  $\mathcal{O}_C(1) - 3[P_0]$  has degree 0, and multiplication by 3 is surjective on  $\text{Pic}^0 C$ .

**Problem 2.6** (Complex multiplication, ★).

Let  $\alpha(X : Y : Z) = (X : Y : \zeta Z)$ , a map defined over  $k(\zeta)$ .

- (a) Show that  $\alpha$  maps  $C_d$  to itself and fixes  $O$ .
- (b) Quote [Sil09, Thm. III.4.8] to conclude that  $\alpha$  is a group homomorphism, hence an automorphism of the elliptic curve  $(C_d, O)$  over  $k(\zeta)$ .
- (c) Show that  $\alpha^2 + \alpha + 1 = 0$  in  $\text{End}_{k(\zeta)}(C_d)$ , the ring of endomorphisms of  $C_d$  defined over  $k(\zeta)$ .<sup>a</sup>
- (d) Deduce that  $\mathbb{Z}[\zeta] \subseteq \text{End}_{k(\zeta)}(C_d)$ , so every Selmer curve has complex multiplication. What does [Sil09, §III.10] then predict for  $j(C_d)$  and for  $\# \text{Aut}_k(C_d)$  and  $\# \text{Aut}_{k(\zeta)}(C_d)$ ?

<sup>a</sup> $\alpha^3 = 1$ , and the endomorphism ring of an elliptic curve has no zero divisors, [Sil09, §III.4].

### 3. FROM RIEMANN–ROCH TO A WEIERSTRASS EQUATION

This section reproduces, for our family, the proof of [Sil09, Prop. III.3.1]: every elliptic curve is isomorphic to a curve in Weierstrass form. You may use the following two facts without proof.

**Fact 3.1** (Bézout). For a nonzero linear form  $h$ , the scheme-theoretic intersection  $h \cap C_d$  is an effective divisor of degree 3 in  $\text{Div } C_d$ ; over  $\bar{k}$  it becomes a sum of three points of  $C_d(\bar{k})$  counted with multiplicity. Consequently, since  $X + Y$  cuts out  $3O$  by Problem 1.2(b),

$$\text{div}\left(\frac{h}{X+Y}\right) = (h \cap C_d) - 3O.$$

**Fact 3.2** (Riemann–Roch for genus one). For every  $n \geq 1$  we have  $\dim_k \mathcal{L}(nO) = n$ , where for  $D \in \text{Div } C_d$

$$\mathcal{L}(D) = \{f \in k(C_d)^\times : \text{div}(f) \geq -D\} \cup \{0\}, \quad \text{div}(f) \in \text{Div } C_d.$$

This is a space of functions defined over  $k$ ; since  $C_d$  is geometrically integral,  $\mathcal{L}(D) \otimes_k \bar{k} = \mathcal{L}(D_{\bar{k}})$ , so the dimension is the same whether computed over  $k$  or over  $\bar{k}$ . See [Sil09, §II.5] and the proof of [Sil09, Prop. III.3.1].

Problem 3.3 is meant to be finished at home. Parts (a)–(c) go quickly once you see that Fact 3.1 converts “pole order at  $O$ ” into “multiplicity of  $O$  in  $h \cap C_d$ ”. Part (e) is a two-line computation with a surprising answer, and (h) is worth bringing to the next session.

**Problem 3.3** (The Weierstrass model of  $C_d$ ).

Consider the rational functions in  $k(C_d)$

$$w := \frac{Z}{X+Y}, \quad v := \frac{X-Y}{X+Y}.$$

The notes use  $X/(X+Y) = (1+v)/2$  in place of  $v$ , which also makes sense in characteristic 2; part (g) recovers that model, which is the one in Proposition 4.8.3 of the notes.

- (a) Using Problem 1.2(a) and Fact 3.1, compute  $\text{div}(w)$ . Deduce that  $w$  has a pole of order exactly 2 at  $O$  and no other pole, so  $w \in \mathcal{L}(2O)$ . Where are its zeros, and what are they in the language of Problem 2.5?
- (b) Show that the line  $X - Y = 0$  does not pass through  $O$ . Deduce that  $v$  has a pole of order exactly 3 at  $O$ , so  $v \in \mathcal{L}(3O)$ . Where are its zeros?
- (c) Using Fact 3.2, explain why  $\{1, w\}$  is a basis of  $\mathcal{L}(2O)$  and  $\{1, w, v\}$  is a basis of  $\mathcal{L}(3O)$ .
- (d) The seven functions  $1, w, v, w^2, wv, w^3, v^2$  all lie in  $\mathcal{L}(6O)$ . Why must they satisfy a non-trivial linear relation over  $k$ ? List their pole orders at  $O$ ; which two collide, and why does that force the relation to involve both?
- (e) Find the relation.<sup>a</sup>
- (f) Assume  $\text{char } k \neq 2$ , and let  $E_1: y_1^2 = x_1^3 - 432d^2$ . Show that the relation of (e) says that there is a morphism  $\varphi_1: C_d \rightarrow E_1$  with  $\varphi_1^*x_1 = 12dw$  and  $\varphi_1^*y_1 = 36dv$ . Write down  $\varphi_1$  and its inverse explicitly.
- (g) Where exactly did you use  $\text{char } k \neq 2$ ? Show that there is instead a morphism  $\varphi_2$  with  $\varphi_2^*x_2 = 3dw$  and  $\varphi_2^*y_2 = 9dX/(X+Y)$  to

$$(3.1) \quad E_2: y_2^2 - 9dy_2 = x_2^3 - 27d^2,$$

which is a Weierstrass equation in the sense of [Sil09, §III.1] valid whenever  $\text{char } k \neq 3$ . Which of  $a_1, \dots, a_6$  are nonzero?

- (h) (★) The relation in (e) has no  $w^2, wv, w$  or  $v$  term, although [Sil09, Prop. III.3.1] allows all four. Explain why, using the automorphisms  $\sigma = \ominus$  of Problem 2.1 and  $\alpha$  of Problem 2.6.<sup>b</sup>

<sup>a</sup>Work on the affine chart  $X+Y=1$ , where  $X=(1+v)/2$  and  $Y=(1-v)/2$ , and expand  $X^3+Y^3$ .

<sup>b</sup>Compute  $\sigma^*w, \sigma^*v, \alpha^*w, \alpha^*v$ ; then use (d) to argue that the relation spans a one-dimensional space, so it is a simultaneous eigenvector.

#### 4. WORKED EXAMPLES IN MAGMA

Most of you have never opened **Magma**, so this section is written as worked examples rather than exercises: read them, then re-run them yourself and change things. Every line of output below is reproduced from **Magma** V2.28-8 [BCP97], with whitespace tidied and one long type name elided. Three short variations to try on your own are collected at the end.

**Warning 4.1** (Closed points in **Magma**). A **closed point**  $P$  of  $C_d$  is a point of the scheme  $C_d$ ; its residue field  $k(P)$  is a finite extension of  $k$ , and  $\deg P := [k(P) : k]$ . The closed points of degree one are the rational points, and a closed point  $P$  with  $k(P)/k$  separable corresponds to an orbit of  $\deg P$  points of  $C_d(\bar{k})$  under  $\text{Gal}(k^{\text{sep}}/k)$ . The divisor group  $\text{Div } C_d$  is the free abelian group on the closed points [Poo06, §1.9], whereas [Sil09, §II.3] works with points of  $C_d(\bar{k})$  and Galois-invariant divisors on  $\bar{C}_d$ . **Magma** follows the first convention: **Places** returns closed points, **Divisor** builds elements of  $\text{Div } C_d$ , and **RiemannRochSpace** returns  $\mathcal{L}(D)$  as a  $k$ -vector space.

Each example continues the previous one; begin with

```
QQ := Rationals();
P2<X,Y,Z> := ProjectiveSpace(QQ,2);
d := 7;
C := Curve(P2, X^3 + Y^3 - d*Z^3);
O := C![1,-1,0];
```

The notation `P2<x,y,z>` names the coordinates so that we can write equations in them, and `c!` coerces a list of three numbers into a rational point of `C`; typing `0 := (1:-1:0);` does not work.

**Example 4.2** (Setting up, and a trap). First, Problem 1.1:

---

```
IsNonSingular(C); // true
Genus(C);         // 1
0;                // (-1 : 1 : 0)
```

---

The last line is worth a pause. We asked for  $(1 : -1 : 0)$  and got  $(-1 : 1 : 0)$ : a point of  $\mathbb{P}^2(\mathbb{Q})$  is defined only up to scaling, and Magma normalises the representative it stores, here by  $-1$ .

Now ask for the Weierstrass model. The second return value `mp` is the isomorphism from `C` to `E`.

---

```
E, mp := EllipticCurve(C, 0);
E;
// Elliptic Curve defined by y^2 - 9/49*y = x^3 - 27/2401 over Rational Field
jInvariant(E); // 0
E eq EllipticCurve([0,0,0,0,-432*d^2]); // false
```

---

That `false` looks alarming, since Problem 3.3(f) says the answer should be  $E_1: y_1^2 = x_1^3 - 432d^2$ . It is not a contradiction: `eq` compares Weierstrass *coefficients*, and by [Sil09, Prop. III.3.1(b)] a Weierstrass model is unique only up to the changes of variables  $x = u^2x' + r$ ,  $y = u^3y' + su^2x' + t$ . The right question is whether the curves are isomorphic, and against which model:

---

```
Eours := EllipticCurve([0,0,-9*d,0,-27*d^2]);
Eours; // y^2 - 63*y = x^3 - 1323
bool, iso := IsIsomorphic(Eours, E);
bool; // true
IsomorphismData(iso); // [ 0, 0, 0, 1/7 ]
```

---

So  $r = s = w = 0$  and  $u = 1/7 = 1/d$ : Magma found exactly the characteristic-free model  $E_2$  of Problem 3.3(g), with  $x_2$  and  $y_2$  divided by  $d^2$  and  $d^3$ . The two maps agree as well,

---

```
DefiningEquations(mp); // [ 3/7*Z, 9/49*X, X + Y ]
```

---

against our  $(3dZ : 9dX : X + Y)$ , the ratios being  $d^2$  and  $d^3$  exactly as  $u = 1/d$  demands. The moral: for elliptic curves, `eq` is almost never the question you mean to ask.

We can even watch this happen for all  $d$  at once, by working over the function field  $\mathbb{Q}(d)$  instead of over  $\mathbb{Q}$ :

---

```
Fd<D> := FunctionField(Rationals());
P2d<Xd,Yd,Zd> := ProjectiveSpace(Fd,2);
Cd := Curve(P2d, Xd^3 + Yd^3 - D*Zd^3);
Ed, mpd := EllipticCurve(Cd, Cd![1,-1,0]);
Ed;
// Elliptic Curve defined by y^2 - 9/D^2*y = x^3 - 27/D^4
DefiningEquations(mpd);
// [ 3/D*Zd, 9/D^2*Xd, Xd + Yd ]
```

---

which is (3.1) rescaled by  $u = d$ , as an identity in  $d$ .

Finally, the predictions of Problem 2.6(d):

---

```
#AutomorphismGroup(E);                // 2
K<zeta> := CyclotomicField(3);
#AutomorphismGroup(BaseChange(E,K));   // 6
```

---

**Example 4.3** (The addition formulas, checked). A Magma function takes arguments and returns a value; here both are sequences of three coordinates, so that (2.3) and (2.4) can be typed almost verbatim.

---

```
dbl := function(P)
  x := P[1]; y := P[2]; z := P[3];
  return [-y*(2*x^3+y^3), x*(x^3+2*y^3), z*(x^3-y^3)];
end function;

add := function(P1,P2)
  A := P1[1]^2*P2[1] + P1[2]^2*P2[2] - d*P1[3]^2*P2[3];
  B := P1[1]*P2[1]^2 + P1[2]*P2[2]^2 - d*P1[3]*P2[3]^2;
  return [B*P1[2]-A*P2[2], B*P1[1]-A*P2[1], B*P1[3]-A*P2[3]];
end function;

P := [2,-1,1];
Q := dbl(P);      C!Q;    // (5/3 : 4/3 : 1)
R := dbl(Q);      C!R;    // (-1256/183 : 1265/183 : 1)
S := add(P,Q);    C!S;    // (-17/38 : 73/38 : 1)
```

---

These are the points of Problem 2.3. Coercing through `c!` normalises them, which is why  $(15 : 12 : 9)$  prints as  $(5/3 : 4/3 : 1)$ . Now transport everything to  $E$  through `mp` and compare with Magma's own group law:

---

```
EP := mp(C!P);
mp(C!Q) eq 2*EP;      // true
mp(C!S) eq 3*EP;      // true
mp(C!R) eq 4*EP;      // true
```

---

Three `true`s: formulas derived from nothing but (2.1) and a  $3 \times 3$  determinant agree with a general-purpose implementation.

**Example 4.4** (Riemann–Roch, mechanically). Magma can carry out Problem 3.3 without being told the answer. The generators `x` and `y` of `FunctionField(C)` are the affine coordinates  $X/Z$  and  $Y/Z$  of  $C_d$  (not Weierstrass coordinates), so our two functions are built from those.

---

```
FF<x,y> := FunctionField(C);    // x = X/Z, y = Y/Z
w := 1/(x+y);                  // Z/(X+Y)
v := (x-y)/(x+y);              // (X-Y)/(X+Y)
p10 := Place(0);
Valuation(w, p10);              // -2
Valuation(v, p10);              // -3
```

---

Those are the pole orders of Problem 3.3(a), (b); a valuation of  $-2$  is a pole of order 2. As in Warning 4.1, `Place(0)` is the closed point  $O$  and `Divisor` builds an element of  $\text{Div } C_d$ , so everything here happens over  $\mathbb{Q}$ . Next, Fact 3.2 itself:

---

```

D := Divisor(p10);
for n in [1..6] do print n, Dimension(RiemannRochSpace(n*D)); end for;
// 1 1
// 2 2
// 3 3
// 4 4
// 5 5
// 6 6

```

---

Now ask for a basis of  $\mathcal{L}(2O)$  without suggesting one:

---

```

V2, m2 := RiemannRochSpace(2*D);
[m2(b) : b in Basis(V2)]; // [ x^2 - y*x + y^2, 1 ]

```

---

which looks nothing like our  $w$ . But  $X^3 + Y^3 = (X + Y)(X^2 - XY + Y^2)$ , and on  $C_d$  the left side is  $dZ^3$ , so

$$\frac{X^2 - XY + Y^2}{Z^2} = \frac{dZ^3}{Z^2(X + Y)} = d \frac{Z}{X + Y} = dw,$$

and indeed

---

```

(x^2 - x*y + y^2) eq d*w; // true

```

---

Finally the relation. We list the seven functions of Problem 3.3(d), express each in a basis of  $\mathcal{L}(6O)$  (that is what @@ does, applying the inverse of the map mV), and take the kernel of the resulting matrix.

---

```

V, mV := RiemannRochSpace(6*D);
fns := [FF!1, w, v, w^2, w*v, w^3, v^2];
[-Valuation(f, p10) : f in fns]; // [ 0, 2, 3, 4, 5, 6, 6 ]
M := Matrix([Eltseq(f @@ mV) : f in fns]);
K := Kernel(M);
Dimension(K); // 1
Basis(K); // [ ( 1 0 0 0 0 -28 3) ]

```

---

The pole orders are 0, 2, 3, 4, 5, 6, 6: seven functions, six available orders, the collision at 6 forcing the relation exactly as in Problem 3.3(d). And the kernel is one-dimensional, spanned by the vector reading

$$1 - 28w^3 + 3v^2 = 0,$$

which is  $3v^2 + 1 = 4dw^3$  at  $d = 7$ . We got it by hand from a factorisation; Magma got it from linear algebra in a Riemann–Roch space.

**Now try these.** Each one is a small edit to Example 4.2 or Example 4.3, and in each case the worksheet already tells you what the answer must be.

- (1) Set  $a := 2$ . Problem 2.1(c) predicts a rational point of order two. Find it, verify with the group law that it has order two, and compute its image under  $\varphi_1$  by hand.
- (2) Set  $a := 8$ . Problem 2.5(g) predicts rational 3-torsion. Find the two rational 3-torsion points other than  $O$ , use your ab1 function to check that  $[2]P = \ominus P$ , and compute  $\varphi_1(P)$  by hand.

- (3) Use `IsIsomorphic` to decide whether  $C_1 \simeq C_8$  and whether  $C_1 \simeq C_2$  over  $\mathbb{Q}$ . Pass it the *Weierstrass* models: on the plane curves themselves `Magma` answers `For genus 0 or 1 curves the`  
 $\hookrightarrow$  `basefield must be finite`. Reconcile the answers with the cubic-twist criterion of Section 5.

## 5. WHERE THE NAME COMES FROM

Selmer [Sel51] studied the general diagonal cubic  $aX^3 + bY^3 + cZ^3 = 0$ , and produced the standard counterexample to the Hasse principle: the curve  $3X^3 + 4Y^3 + 5Z^3 = 0$  has real points and  $\mathbb{Q}_p$ -points for every prime  $p$ , but no rational point. Such a curve is a torsor under its Jacobian [Poo17, §5.12], and that Jacobian is a Selmer curve in our sense:

$$\text{Jac}(aX^3 + bY^3 + cZ^3) \simeq (y^2 = x^3 - 432(abc)^2).$$

Two ways to test this in `Magma`: the curve  $X^3 + 2Y^3 + 3Z^3 = 0$  *does* have the rational point  $(1 : 1 : -1)$ , so it is its own Jacobian and should be isomorphic to  $C_6$ , that is to  $y^2 = x^3 - 432 \cdot 6^2$ ; and for Selmer's own curve one can call `Jacobian(GenusOneModel(3*X^3 + 4*Y^3 + 5*Z^3))` and compare with  $y^2 = x^3 - 432 \cdot 60^2$ . Both come out `true`.

Finally, by Problem 3.3(f) the curves  $C_d$  and  $C_{d'}$  are isomorphic over  $k$  exactly when  $d/d' \in (k^\times)^3$ , so the family is parametrised by  $k^\times/(k^\times)^3$ . These are the **cubic twists** of  $X^3 + Y^3 = Z^3$ , and they are the natural first examples for the descent machinery of [Sil09, §X] and [Poo17, Ch. 8].

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